B020312(020)

B. Tech. (Third Semester) Examination, Nov.-Dec. 2022

(AICTE Scheme)

(Mechanical Engineering Branch)

INTRODUCTION to FLUID MECHANICS

Time Allowed : Three hours

Maximum Marks : 100

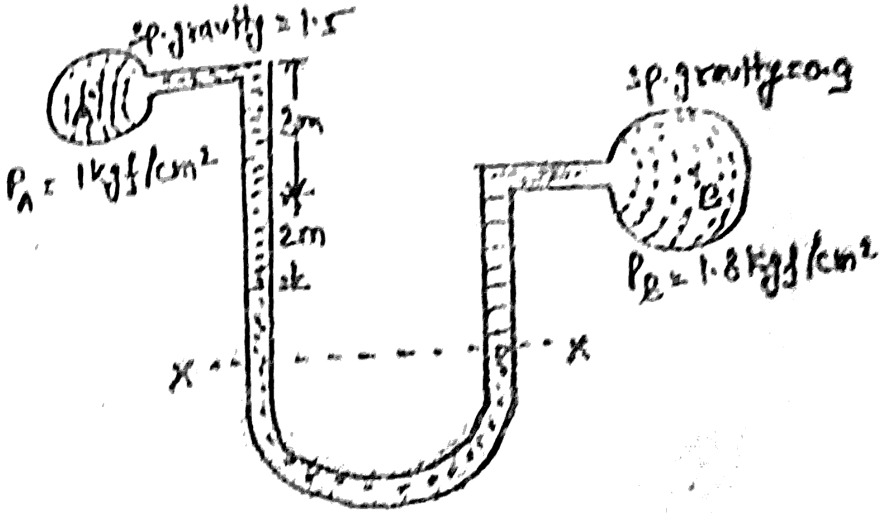
Minimum Pass Marks : 35

Note : *Attempt all questions. Part (a) of each question is compulsory and carries 4 marks each. Attempt any two parts from (b), (c) and (d) of each question and carries 8 marks each.*

Unit-I

1. (a) What are the difference between dynamic viscosity & kinematic viscosity?

(b) A differential manometer is connected at the two points A and B of two pipes as shown in figure. The pipe A contains a liquid of sp. gravity = 1·5, while pipe B contains a liquid of sp. gravity = 0·9. The pressure at A and B are 1 kgf/cm^2 and 1·80 kgf/cm^2 respectively. Find the difference in mercury level in the differential manometer.



(c) 3·2 m^3 of a certain oil has weight 27·5 kN. Calculate its specific weight, mass density, specific volume and specific gravity with respect to water if kinematic viscosity of the oil is 7×10^{-3} stoke, what would be its dynamic viscosity in centipoise?

(d) Explain stability of floating body with help of metacentric height and position of centre of gravity of body.

Unit-II

2. (a) Compare path live and stream line.

(b) For the following flows, find the equation of the stream lines passing through (2, 2).

(i) $V = 3x\hat{i} - 3y\hat{j}$

(ii) $V = -y^2\hat{i} - 6x\hat{j}$

(c) Derive continuity equation of liquid flow based on conservation of mass prinicple.

(d) The stream function for a two dimensional flow is given by $\psi = 2xy$, calculate the velocity at the point $P(2,3)$. Find the velocity potential function ϕ.

Unit-III

3. (a) Explain impulse-momentum equation.

(b) The water is flowing through a pipe having diameters 20 cm and 10 cm at section 1 and 2 respectively. The rate of flow through pipe is 35 litre/s. The section 1 is 6 m above datum and section 2 is 4 above datum, If the pressure at section 1 is 39.24 N/cm^2